A Modified Murray's Law for Insect Respiration

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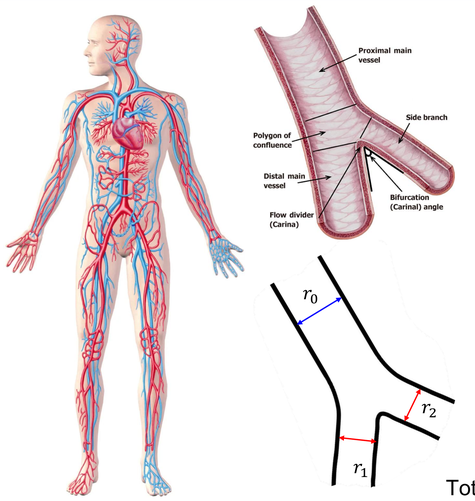
Classical Murray's Law



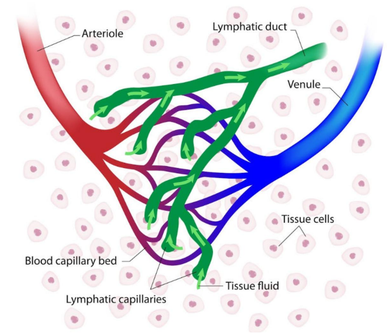
- Biological branching networks such as blood vessels, respiratory airways, and plant veins must transport oxygen, nutrients, or other materials efficiently while using as little energy as possible.
- **Murray's law predicts how these networks branch to achieve optimal transport.**
- This principle can be extended beyond biology to the design of efficient microscale transport networks, such as microfluidic channels and lab-on-a-chip systems.

$$\mathcal{J}(r) = \frac{8\mu L}{\pi r^4} Q^2 + \pi r^2 L \sigma \xrightarrow{\text{Minimize}} r_p^k = r_{d1}^k + r_{d2}^k + \cdots + r_{dn}^k, \quad k = 3$$

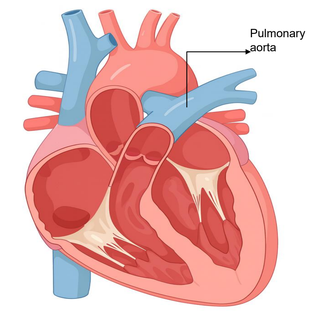
Total power required — Viscous dissipation — Metabolic upkeep cost — Original Murray's exponent

Image sources: www.istockphoto.com, http://dx.doi.org/10.4330/wjc.v14.i3.108

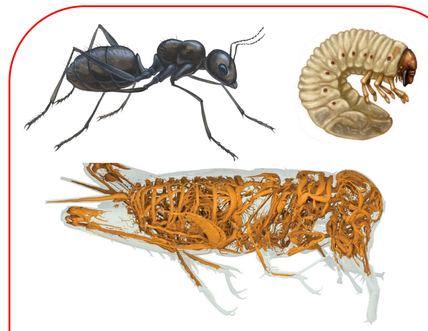
Biological Systems NOT Following Murray's Law



(a) Lymphatic system ($k \approx 1.45$, Talkington et al. 2022)



(b) Pulsating pulmonary aorta ($k \approx 2.3$ for Correa et al. 2025)

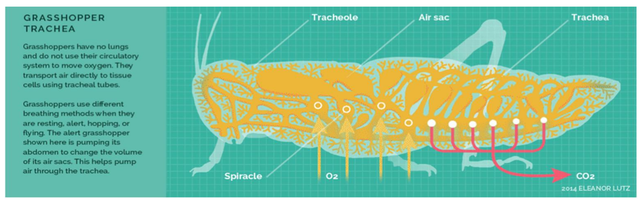


(c) Insect respiratory system ($k \approx 1$, Aitkenhead et al. 2020, $k \approx 1$, Nunome 1951, $k \approx 2$, Krogh 1920, Locke 1958, $k < 3$, Socha et al., 2010)

Why do insect respiratory networks break Murray's law?

Image sources: (a) https://theskinstudios.com.au (b) https://www.biorender.com (c) https://cdn.britannica.com and https://www.amnh.org/

Insect Features



Source: Eleanor Lutz, https://tabletopwhale.com/2014/10/24/3-different-ways-to-breathe.html.

Feature Set 1
A. Diffusive transport is involved
B. Hydrodynamic slip is present
C. Metabolic upkeep cost is different

Feature Set 2
A. Tidal transport
B. Source driven CO2 exhalation
C. Knudsen effect

Modified Murray's Formulation

$$\mathcal{J}(r) = \frac{8\mu L}{\pi r^4 \left(1 + \frac{4\lambda}{r}\right)} Q_a^2 + \frac{R'TCL}{\pi r^2 D} Q_d^2 + B r^p L$$

- p: Metabolic upkeep exponent, $p = 1$: area proportional; $p = 2$: volume proportional
- $\frac{R'TCL}{\pi r^2 D} Q_d^2$: Diffusive power dissipation
- $\frac{8\mu L}{\pi r^4 (1 + \frac{4\lambda}{r})} Q_a^2$: Viscous power dissipation
- $\left(1 + \frac{4\lambda}{r}\right)$: Slip correction factor

$$\mathcal{J}(r) = \sum_j \theta_j A_j L X_j^2 r^{-m_j} + B L r^p$$

[A duty factor, θ, is assigned to each phase]

$$k = \frac{\hat{m} + p}{2}$$

Where, $\hat{m}(r) = \frac{\sum_j m_j \psi_j(r)}{\sum_j \psi_j(r)}$ — Weighted contribution of both transport modes In modified exponent k

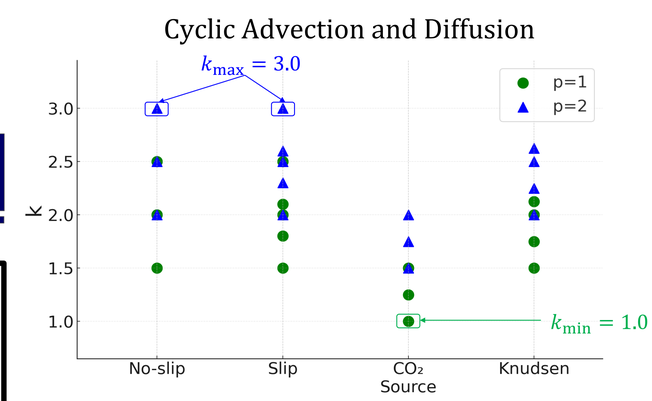
Modified formulation:
- Model 1: Includes feature set 1
- Model 2: Includes feature set 1+2

Model 1 Predictions

Model	Limit	Regime	k	
			$p = 1$	$p = 2$
Model 1: Simultaneous Adv + Diff	$Pe \ll 1$	Diffusive	1.5	2.0
	$Pe \gg 1$	Advective	2.5	3.0
	$Pe \approx 1$	Balanced	$[1.5 - 2.5]$	$[2.0 - 3.0]$

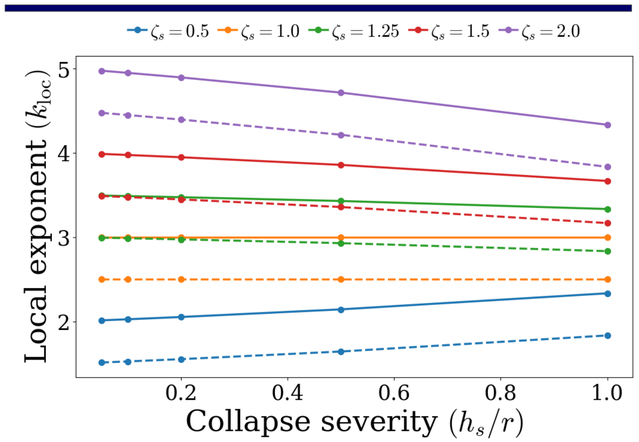
The simultaneous advection & diffusion model with slip shows that branching exponent k varies with the Péclet number, Pe (Pe: advection-diffusion ratio).

Model 2 Predictions



The cyclic advection & diffusion model incorporating insect-specific physics shows that the branching exponent k spans a wider range due to CO_2-driven wall flux and Knudsen diffusion effects.

Local k During Collapse



Local branching exponent as a function of collapse severity different collapse scaling exponents ζs, where ζs quantifies how the collapse thickness scales with tube radius. dashed and solid lines correspond to p=1 and p=2, respectively.

Summary

- A **modified Murray's law** explains deviations from the classical cubic scaling ($k = 3$) in insect respiration
- **Model 1** (Simultaneous advection + diffusion) incorporates key microscale transport mechanisms absent in the original formulation, including slip flow and diffusive transport. It predicts $k \approx 1.5$–2.5 (p=1) and $k \approx 2.0$–3.0 (p=2), spanning diffusive to advective limits.
- **Model 2** (Cyclic advection + diffusion) captures insect-specific respiratory physics, including CO_2-driven advective flushing from the tracheal wall and Knudsen diffusion effects. It predicts $k \approx 1.0$–3.0, consistent with observed deviations in biological systems.
- At collapsed sites, the local exponent can exceed $k > 3$, depending on collapse severity (h_s/r)and scaling (ζ_s).

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Acknowledgements

References